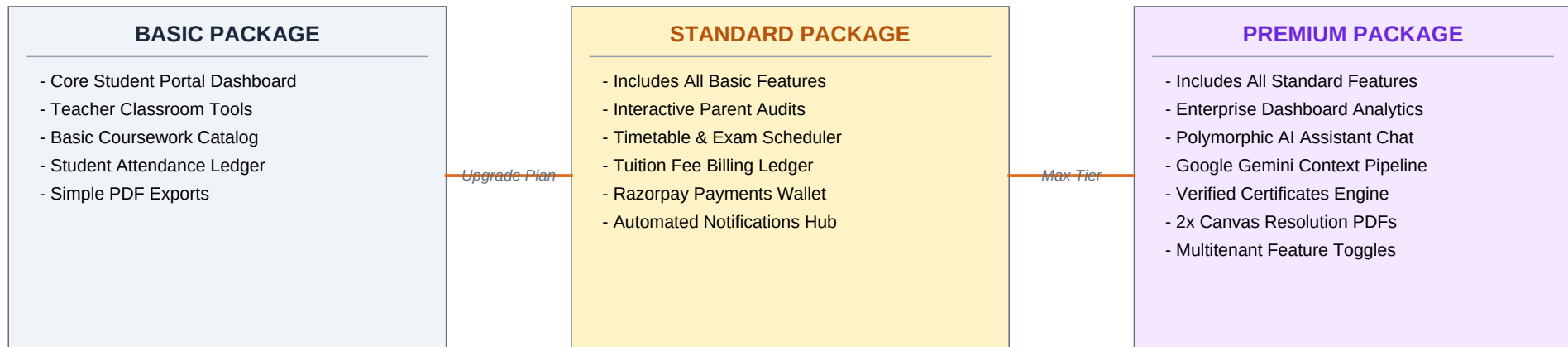


ZENELAIT LMS - COMPREHENSIVE PLATFORM MANUAL & FLOWCHART

Role-Based Capabilities & Package-Wise Feature Access Distribution

1. Package-Wise Feature Access Distribution

Zenelait LMS features are grouped into subscription tiers (Basic, Standard, Premium) managed at the platform level by the Ultra Super Admin. Organizations subscribe to these packages to grant access to corresponding features across their active users.



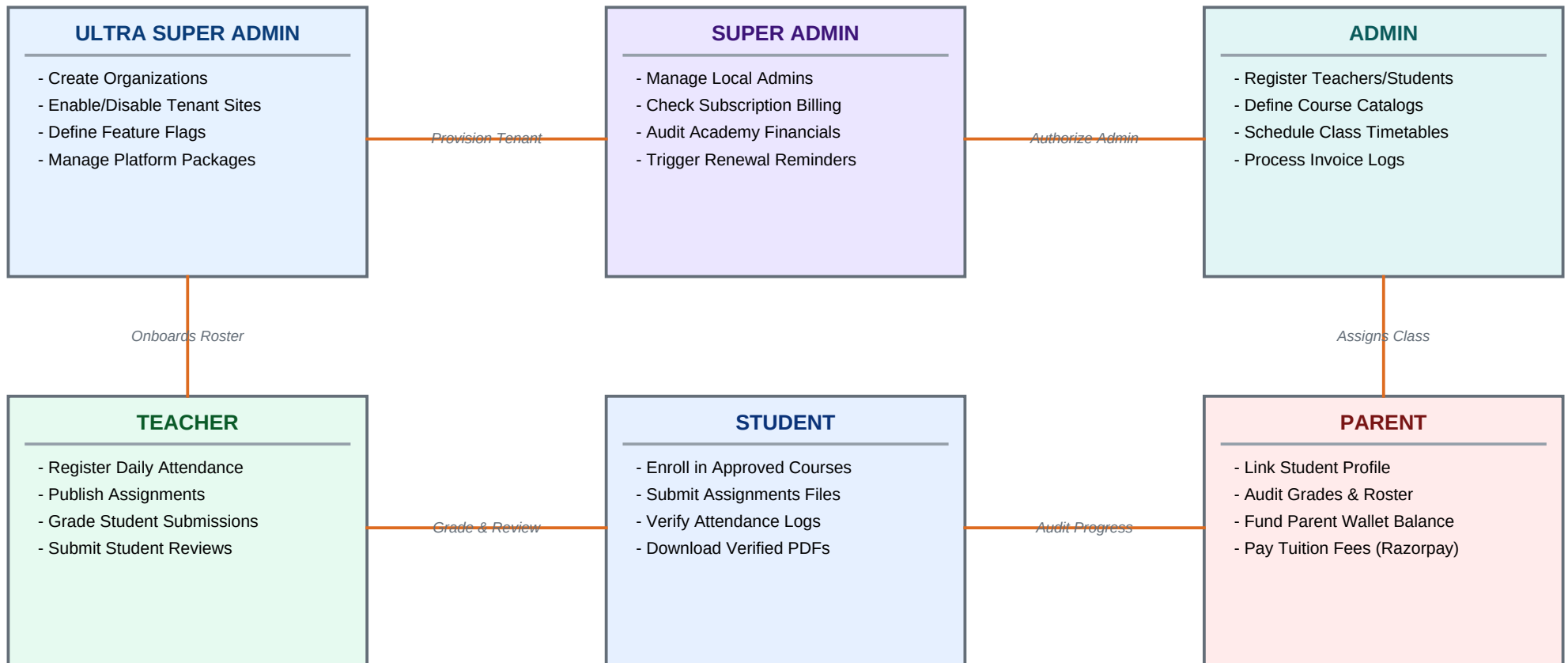
PACKAGE MANAGEMENT RULES & ARCHITECTURE:

1. Active subscription package levels dictate API availability across all role endpoints dynamically.
2. If an organization's subscription expires, all write operations and role access guards block dashboard access.
3. Premium certificate generation utilizes html2canvas rendering at 2x scale and jsPDF to compile verified achievements.
4. AI Assistant uses gemini-2.5-flash with customized context mapping according to the logged-in user's role.

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2. Role-Based Permissions & Platform Integration Flowchart



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3. System Workflow Sequence (Data & Transactions)

The following list outlines the step-by-step transaction flow of Zenelait LMS from system initialization to certified student graduation:

- Step 1: Tenant Provisioning -->** Ultra Super Admin registers the Organization (Academy) and toggles active global features. Local database resources are mapped to the cluster.
- Step 2: Subscription Activation -->** Super Admin selects a package (Basic/Standard/Premium). Razorpay Payment Gateway completes subscription transaction and activates API endpoints.
- Step 3: User Onboarding -->** Super Admin creates local Admins. Admins onboard Teachers, Students, and Parents. Parents link their profile to their student's registration code.
- Step 4: Academic Setup -->** Admins create Departments, Courses, Batches, and Class Timetables. Timetables are exposed via endpoints protecting GET routes for Teachers and Students.
- Step 5: Classroom Execution -->** Teachers log student attendance and publish assignments. System schedules execute dynamic status updates across batches automatically on cron schedule.
- Step 6: Audits & Finance -->** Parents inspect their student's progress and pay invoices via the Razorpay parent wallet. Super Admin verifies revenue flows in local reporting graphs.
- Step 7: AI Assistant Support -->** Users interact with the embedded AI chatbot. The chatbot queries SQL views to provide context-aware insights powered by the Gemini 2.5 API.
- Step 8: Completion & Certification -->** Upon course completion, Admin issues academic certificates. Student reviews and prints the certificate with a high-resolution, verified PDF download.